Fleet condition report: Penmanshiel wind farm

Penmanshiel wind farm (14x Senvion MM82, UK), Cubico, Greenbyte export. 14 turbines, SCADA 02 Jun 2016 to 31 Dec 2024; assessment window: the last 24 months of each turbine (Jan 2022 to Dec 2024). Generated by `windml report` from the mapping `penmanshiel_all_years.yaml`.

Executive summary

12 of 14 turbines need attention (8 red, 4 amber). The rest show no warning sign in the data we have.

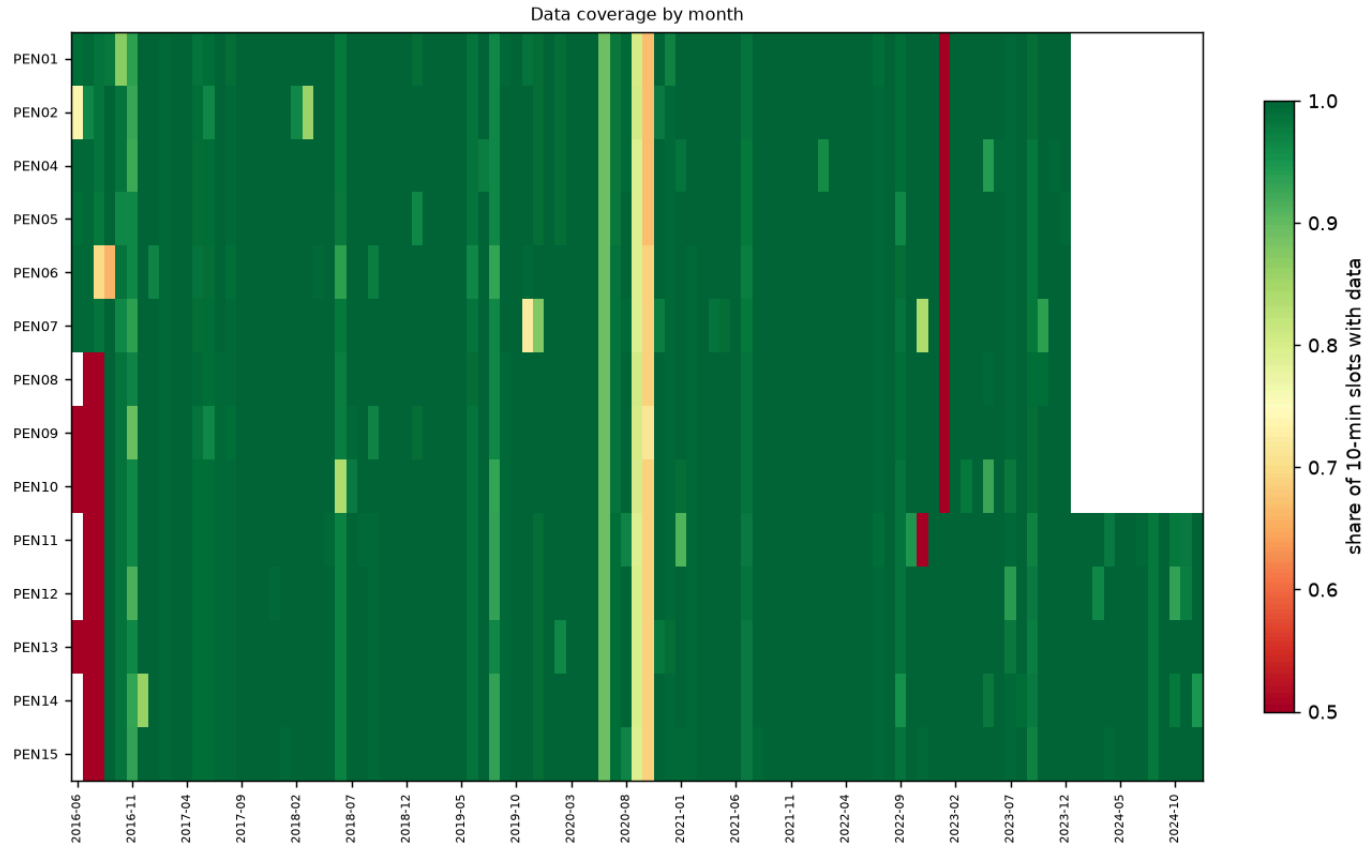
Turbine	Flag	Why	Confidence	Suggested next step
PEN15	RED	It tripped on oil-system faults in 16 separate episodes in two years; it was down for faults or maintenance 3.0% of the time; it tripped on pitch faults in 5 separate episodes in two years; and more (see scorecard).	High	Inspect the oil circuit (filter, pump, cooler, level); review the fault history below and the open work orders.
PEN04	RED	It was down for faults or maintenance 5.8% of the time; it tripped on oil-system faults in 11 separate episodes in two years; it behaved abnormally in the gearbox oil system on 56% of days last year; and more (see scorecard).	High	Review the fault history below and the open work orders; inspect the oil circuit (filter, pump, cooler, level).
PEN06	RED	It was down for faults or maintenance 5.5% of the time; it tripped on oil-system faults in 4 separate episodes in two years; it tripped on pitch faults in 5 separate episodes in two years; and more (see scorecard).	High	Review the fault history below and the open work orders; inspect the oil circuit (filter, pump, cooler, level).
PEN02	RED	It tripped on oil-system faults in 13 separate episodes in two years; it behaved abnormally in the gearbox oil system on 57% of days last year; it has 4.4% of its SCADA data missing.	High	Inspect the oil circuit (filter, pump, cooler, level); trend oil temperature and pressure against the sister turbines; take an oil sample.
PEN01	RED	It behaved abnormally in the gearbox oil system on 38% of days last year; it has 4.4% of its SCADA data missing.	High	Trend oil temperature and pressure against the sister turbines; take an oil sample; ask the scada provider about the gaps (communication or logger problems hide faults).
PEN10	RED	It was down for faults or maintenance 8.9% of the time; it has 4.9% of its SCADA data missing.	High	Review the fault history below and the open work orders; ask the scada provider about the gaps (communication or logger problems hide faults).
PEN11	RED	It behaved abnormally in the gearbox oil system on 42% of days last year; it has a wind speed sensor that drifted -5% away from the other turbines.	High	Trend oil temperature and pressure against the sister turbines; take an oil sample; recalibrate or replace the sensor (models and power-curve checks rely on it).
PEN14	RED	It needs 2.1 bar more gear-oil pump pressure at full load than earlier in its life.	High	Check the oil filter differential pressure and cooler; take an oil sample.
PEN07	AMBER	It tripped on oil-system faults in 5 separate episodes in two years; it tripped on pitch faults in 4 separate episodes in two years; it behaved abnormally in the gearbox oil system on 14% of days last year; and more (see scorecard).	High	Inspect the oil circuit (filter, pump, cooler, level); inspect pitch drives, batteries and slip ring.
PEN05	AMBER	It was down for faults or maintenance 3.4% of the time; it behaved abnormally in the gearbox oil system on 12% of days last year; it has 4.6% of its SCADA data missing.	High	Review the fault history below and the open work orders; trend oil temperature and pressure against the sister turbines; take an oil sample.
PEN08	AMBER	It tripped on oil-system faults in 6 separate episodes in two years; it has 4.5% of its SCADA data missing.	High	Inspect the oil circuit (filter, pump, cooler, level); ask the scada provider about the gaps (communication or logger problems hide faults).
PEN09	AMBER	It tripped on oil-system faults in 7 separate episodes in two years; it has 4.4% of its SCADA data missing.	High	Inspect the oil circuit (filter, pump, cooler, level); ask the scada provider about the gaps (communication or logger problems hide faults).

No action needed now: PEN12, PEN13.

How sure are we. Confidence is *High* when the turbine has 3+ years of history, near-complete data, models trained on earlier years and an alarm log; it drops when any of these is missing. A red flag is a reason to look, not a diagnosis.

1. Data quality

Turbine	First data	Last data	Months	Coverage (window)	Longest gap (h)	Signals	Native interval	Frozen rows (window)	Stuck sensors (>= 1%)	Stop/fault labels
PEN01	2016-06-06	2023-12-31	91	95.6%	744	9	10min	0	-	alarm log
PEN02	2016-06-03	2023-12-31	91	95.6%	744	9	10min	0	-	alarm log
PEN04	2016-06-13	2023-12-31	91	95.1%	744	9	10min	661	-	alarm log
PEN05	2016-06-15	2023-12-31	90	95.4%	744	9	10min	0	-	alarm log
PEN06	2016-06-02	2023-12-31	91	95.5%	744	9	10min	0	-	alarm log
PEN07	2016-06-02	2023-12-31	91	94.6%	744	9	10min	0	-	alarm log
PEN08	2016-07-27	2023-12-31	89	95.5%	744	9	10min	0	-	alarm log
PEN09	2016-06-24	2023-12-31	90	95.6%	744	9	10min	0	-	alarm log
PEN10	2016-06-27	2023-12-31	90	95.1%	744	9	10min	0	-	alarm log
PEN11	2016-07-19	2024-12-31	101	99.4%	16	9	10min	0	-	alarm log
PEN12	2016-07-02	2024-12-31	102	99.0%	50	9	10min	2	-	alarm log
PEN13	2016-06-30	2024-12-31	102	99.7%	10	9	10min	0	-	alarm log
PEN14	2016-07-09	2024-12-31	102	99.4%	22	9	10min	0	-	alarm log
PEN15	2016-07-14	2024-12-31	102	99.7%	9	9	10min	1,233	-	alarm log



Signals present in at least one turbine: `ambient_temp_c`, `available_capacity_kw`, `converter_inlet_temp_c`, `data_availability`, `energy_export_kwh`, `gear_oil_inlet_pressure_bar`, `gear_oil_inlet_temp_c`, `gear_oil_pump_pressure_bar`, `gear_oil_temp_c`, `gen_bearing_front_temp_c`, `gen_bearing_rear_temp_c`, `generator_speed_rpm`, `hub_temp_c`, `lost_prod_curtail_total_kwh`, `lost_prod_downtime_kwh`, `main_bearing_front_temp_c`, `main_bearing_rear_temp_c`, `nacelle_position_deg`, `nacelle_temp_c`, `pitch_a_deg`, `pitch_b_deg`, `pitch_c_deg`, `pitch_motor_temp_1_c`, `pitch_motor_temp_2_c`, `pitch_motor_temp_3_c`, `power_kw`, `power_setpoint_kw`, `rotor_bearing_temp_c`, `rotor_speed_rpm`, `wind_direction_deg`, `wind_speed_1_ms`, `wind_speed_2_ms`, `wind_speed_ms`, `yaw_bearing_angle_deg`.

2. Turbine scorecard

Turbine	Flag	Confidence	Tech. avail.	Energy avail.	Fault h/yr	Largest fault cause	Oil ep.	Pitch ep.	Abnormal oil days, final yr	Under-perf. days, final yr	Pump pressure: base to last (bar)	Oil-temp model error
PEN01	RED	High	99.1%	97.6%	39	pitch (76 h)	2	3	38%	0%	7.2 to 7.5	1.1x
PEN02	RED	High	98.5%	97.5%	92	gearbox (108 h)	13	0	57%	0%	7.7 to 8.1	1.1x
PEN04	RED	High	94.2%	94.6%	262	unspecified (375 h)	11	1	56%	0%	7.7 to 8.0	0.9x
PEN05	AMBER	High	96.6%	94.6%	166	pitch (251 h)	1	1	12%	0%	7.8 to 8.0	0.9x
PEN06	RED	High	94.5%	91.9%	143	pitch (257 h)	4	5	3%	0%	7.5 to 7.7	1.0x
PEN07	AMBER	High	97.4%	95.4%	183	hydraulics (164 h)	5	4	14%	0%	7.0 to 7.8	1.2x
PEN08	AMBER	High	98.0%	96.5%	114	converter (154 h)	6	2	0%	0%	6.8 to 6.9	0.9x
PEN09	AMBER	High	98.6%	97.3%	87	gearbox (119 h)	7	0	1%	0%	6.8 to 6.9	0.9x
PEN10	RED	High	91.1%	92.4%	739	unspecified (1471 h)	0	3	0%	0%	6.7 to 7.0	0.9x
PEN11	RED	High	98.5%	95.7%	66	unspecified (37 h)	1	1	42%	0%	6.8 to 6.6	1.1x
PEN12	GREEN	High	98.0%	95.4%	116	pitch (77 h)	3	3	1%	0%	7.1 to 7.6	1.1x
PEN13	GREEN	High	99.5%	97.5%	6	tower (11 h)	0	0	1%	0%	7.3 to 7.6	1.2x
PEN14	RED	High	98.9%	97.3%	38	electrical (70 h)	0	0	8%	0%	7.8 to 9.9	1.4x
PEN15	RED	High	97.0%	95.4%	89	pitch (71 h)	16	5	22%	1%	7.4 to 6.9	3.1x

Flags and reasons

- **PEN01** RED (confidence High: 91 months of history, models trained on earlier years, several independent indicators agree). RED: abnormal oil state 38% of final-year days; data missing 4.4%
- **PEN02** RED (confidence High: 91 months of history, models trained on earlier years, several independent indicators agree). RED: 13 oil-system trip episodes; RED: abnormal oil state 57% of final-year days; data missing 4.4%
- **PEN04** RED (confidence High: 91 months of history, models trained on earlier years, several independent indicators agree). RED: technical availability 94.2%; RED: 11 oil-system trip episodes; RED: abnormal oil state 56% of final-year days; data missing 4.9%
- **PEN05** AMBER (confidence High: 90 months of history, models trained on earlier years, several independent indicators agree). technical availability 96.6%; abnormal oil state 12% of final-year days; data missing 4.6%
- **PEN06** RED (confidence High: 91 months of history, models trained on earlier years, several independent indicators agree). RED: technical availability 94.5%; 4 oil-system trip episodes; 5 pitch trip episodes; data missing 4.5%
- **PEN07** AMBER (confidence High: 91 months of history, only 95% data coverage, models trained on earlier years, several independent indicators agree). 5 oil-system trip episodes; 4 pitch trip episodes; abnormal oil state 14% of final-year days; data missing 5.4%
- **PEN08** AMBER (confidence High: 89 months of history, models trained on earlier years, several independent indicators agree). 6 oil-system trip episodes; data missing 4.5%
- **PEN09** AMBER (confidence High: 90 months of history, models trained on earlier years, several independent indicators agree). 7 oil-system trip episodes; data missing 4.4%
- **PEN10** RED (confidence High: 90 months of history, models trained on earlier years, several independent indicators agree). RED: technical availability 91.1%; data missing 4.9%

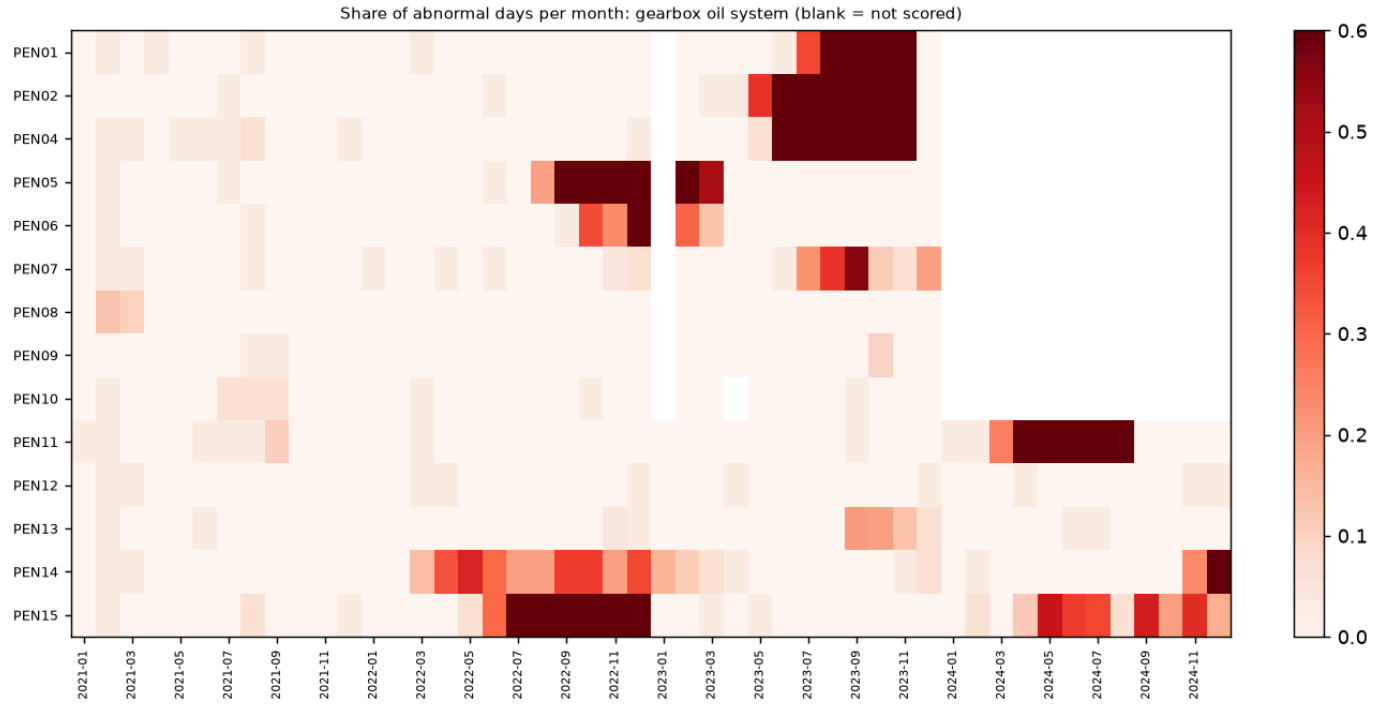
- **PEN11** RED (confidence High: 101 months of history, models trained on earlier years, several independent indicators agree). RED: abnormal oil state 42% of final-year days; wind speed sensor drift -5% vs fleet
- **PEN12** GREEN (confidence High: 102 months of history, models trained on earlier years). No trigger.
- **PEN13** GREEN (confidence High: 102 months of history, models trained on earlier years). No trigger.
- **PEN14** RED (confidence High: 102 months of history, models trained on earlier years). RED: pump pressure +2.1 bar vs baseline
- **PEN15** RED (confidence High: 102 months of history, models trained on earlier years, several independent indicators agree). RED: 16 oil-system trip episodes; technical availability 97.0%; 5 pitch trip episodes; abnormal oil state 22% of final-year days; oil-temp model error 3.1x baseline; wind speed sensor drift -5% vs fleet

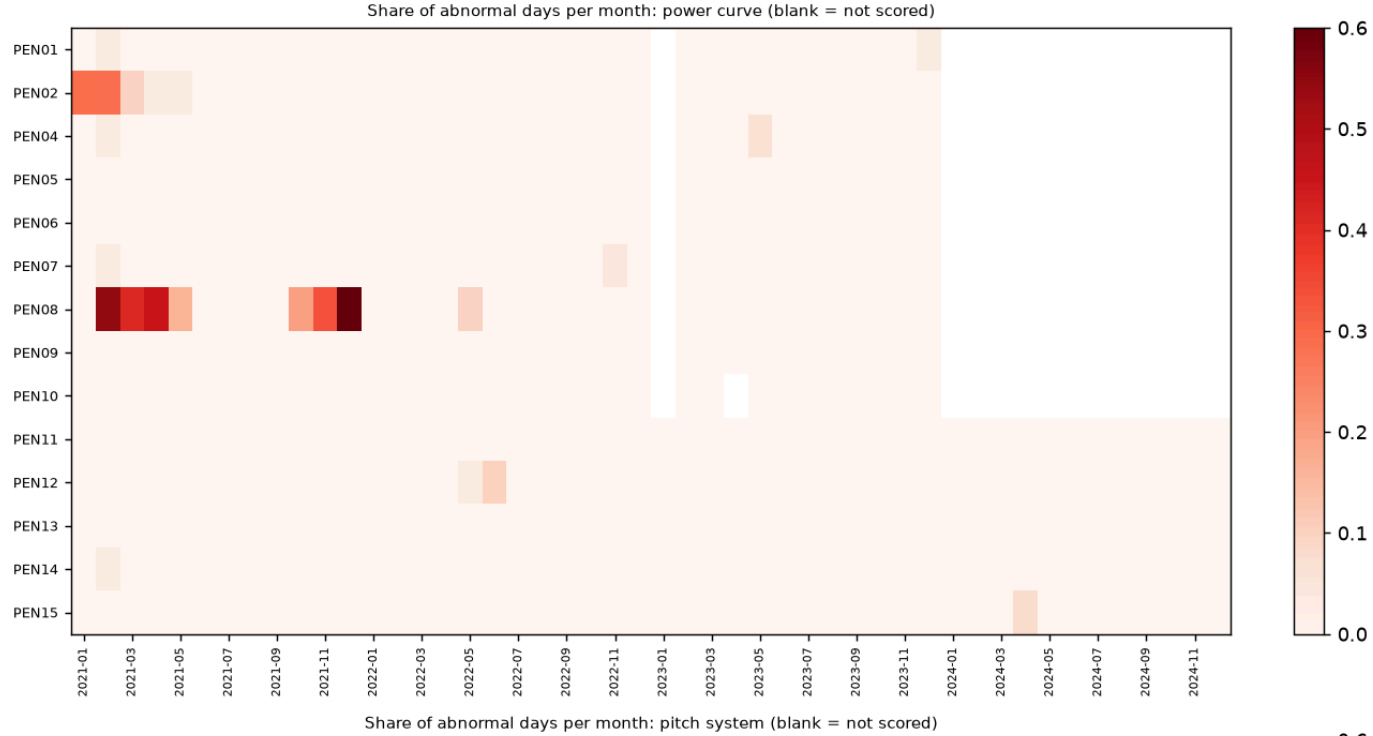
3. Model-based health (normal-behaviour residuals)

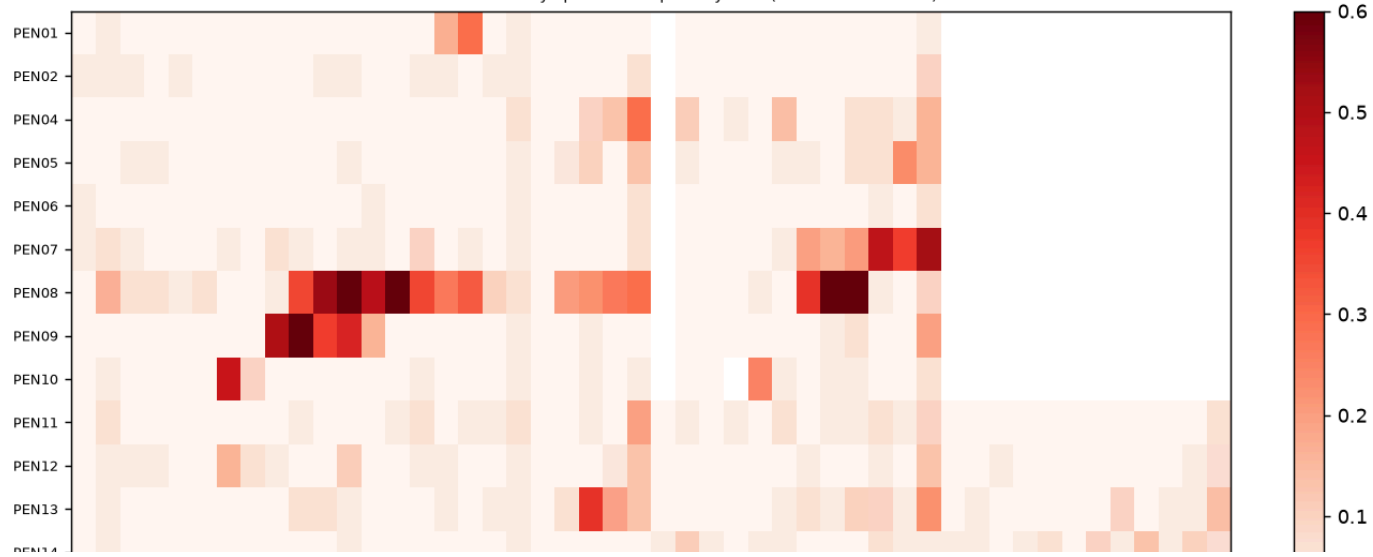
Each turbine gets its own models that predict a signal (oil temperature, bearing temperature, pitch angle, power...) from operating conditions. A day is *abnormal* when the turbine's median deviation from its own model is beyond 3 robust standard deviations in the harmful direction. *Chronic* means at least 30% abnormal days in a month.

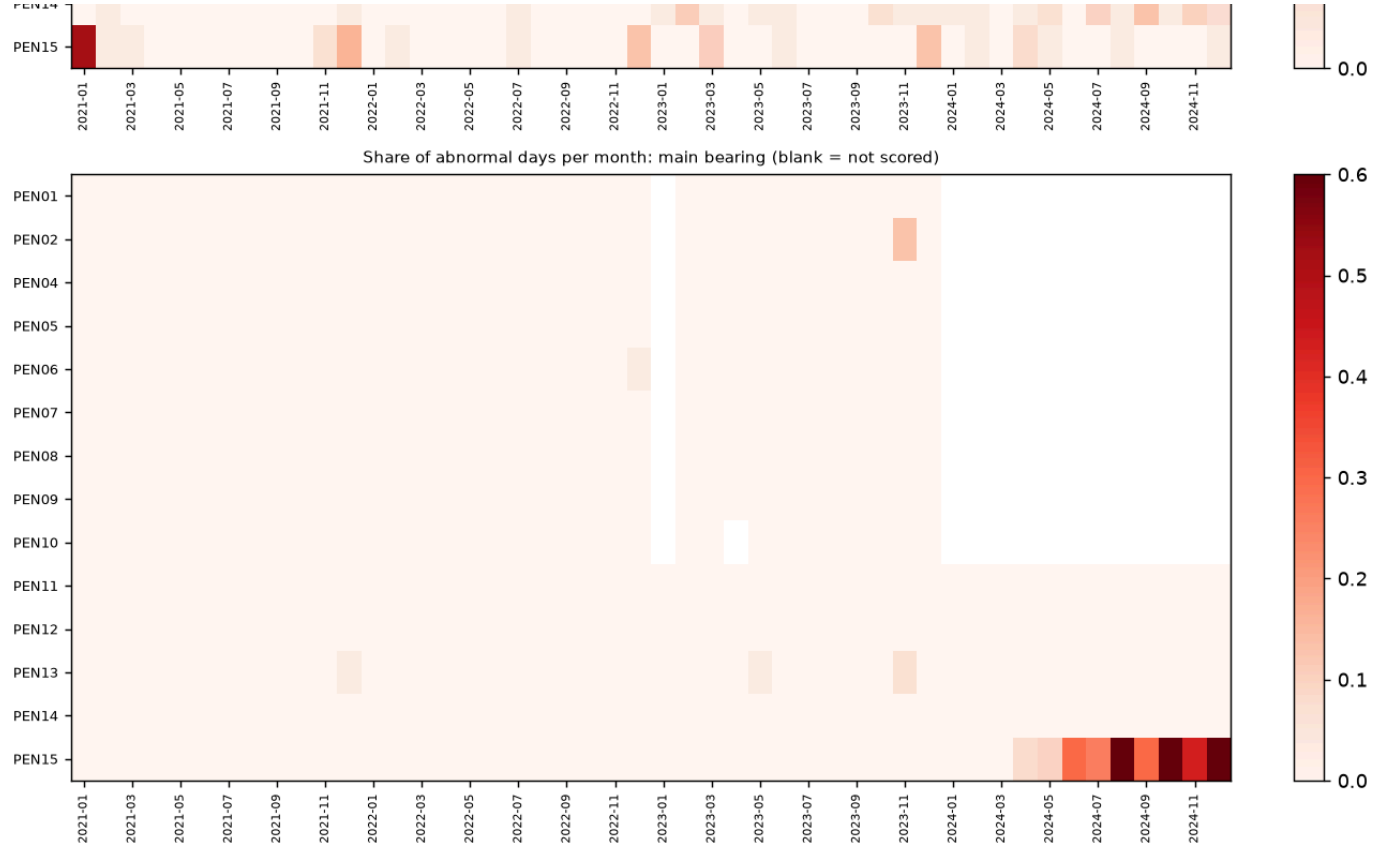
Turbine	Training scheme	Models	Window scored	gearbox oil system: abnormal days final yr / chronic months	main bearing: abnormal days final yr / chronic months	pitch system: abnormal days final yr / chronic months	power curve: abnormal days final yr / chronic months
PEN01	walk-forward	14	95%	38% / 5	0% / 0	0% / 0	0% / 0
PEN02	walk-forward	14	94%	57% / 7	1% / 0	1% / 0	0% / 0
PEN04	walk-forward	14	90%	56% / 6	0% / 0	6% / 0	0% / 0
PEN05	walk-forward	14	92%	12% / 6	0% / 0	6% / 0	0% / 0
PEN06	walk-forward	14	91%	3% / 3	0% / 0	1% / 0	0% / 0
PEN07	walk-forward	14	94%	14% / 2	0% / 0	18% / 3	0% / 0
PEN08	walk-forward	14	95%	0% / 0	0% / 0	18% / 7	0% / 0
PEN09	walk-forward	14	95%	1% / 0	0% / 0	3% / 0	0% / 0
PEN10	walk-forward	14	87%	0% / 0	0% / 0	3% / 0	0% / 0
PEN11	walk-forward	14	99%	42% / 5	0% / 0	1% / 0	0% / 0
PEN12	walk-forward	14	99%	1% / 0	0% / 0	1% / 0	0% / 0
PEN13	walk-forward	14	100%	1% / 0	0% / 0	3% / 0	0% / 0
PEN14	walk-forward	14	99%	8% / 1	0% / 0	5% / 0	0% / 0
PEN15	walk-forward	14	98%	22% / 5	29% / 6	2% / 0	1% / 0

Chronic abnormal turbines (3+ consecutive months with >= 30% abnormal days): PEN01 gearbox oil system (5 months); PEN02 gearbox oil system (7 months); PEN04 gearbox oil system (6 months); PEN05 gearbox oil system (6 months); PEN07 pitch system (3 months); PEN08 pitch system (3 months); PEN11 gearbox oil system (5 months); PEN15 gearbox oil system (3 months); PEN15 main bearing (5 months).


Share of abnormal days per month: gearbox oil system (blank = not scored)
PEN01
PEN02
PEN04
PEN05
PEN06
PEN07
PEN08
PEN09
PEN10
PEN11
PEN12
PEN13
PEN14
PEN15
2021-01
2021-03
2021-05
2021-07
2021-09
2021-11
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0

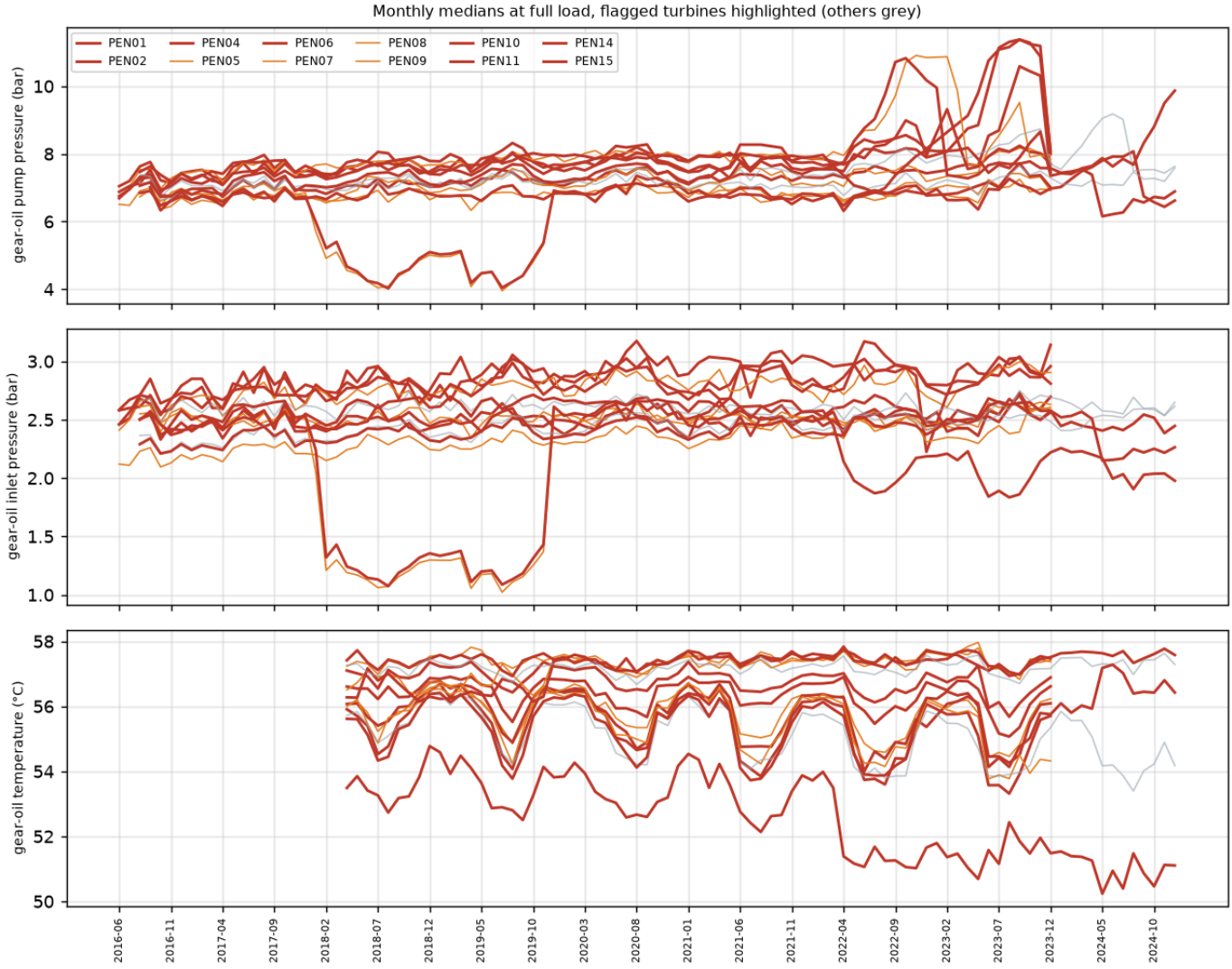

Share of abnormal days per month: power curve (blank = not scored)
PEN01
PEN02
PEN04
PEN05
PEN06
PEN07
PEN08
PEN09
PEN10
PEN11
PEN12
PEN13
PEN14
PEN15
2021-01
2021-03
2021-05
2021-07
2021-09
2021-11
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0


Share of abnormal days per month: pitch system (blank = not scored)
PEN01
PEN02
PEN04
PEN05
PEN06
PEN07
PEN08
PEN09
PEN10
PEN11
PEN12
PEN13
0.6
0.5
0.4
0.3
0.2
0.1



4. Gearbox oil pressure and temperature trends

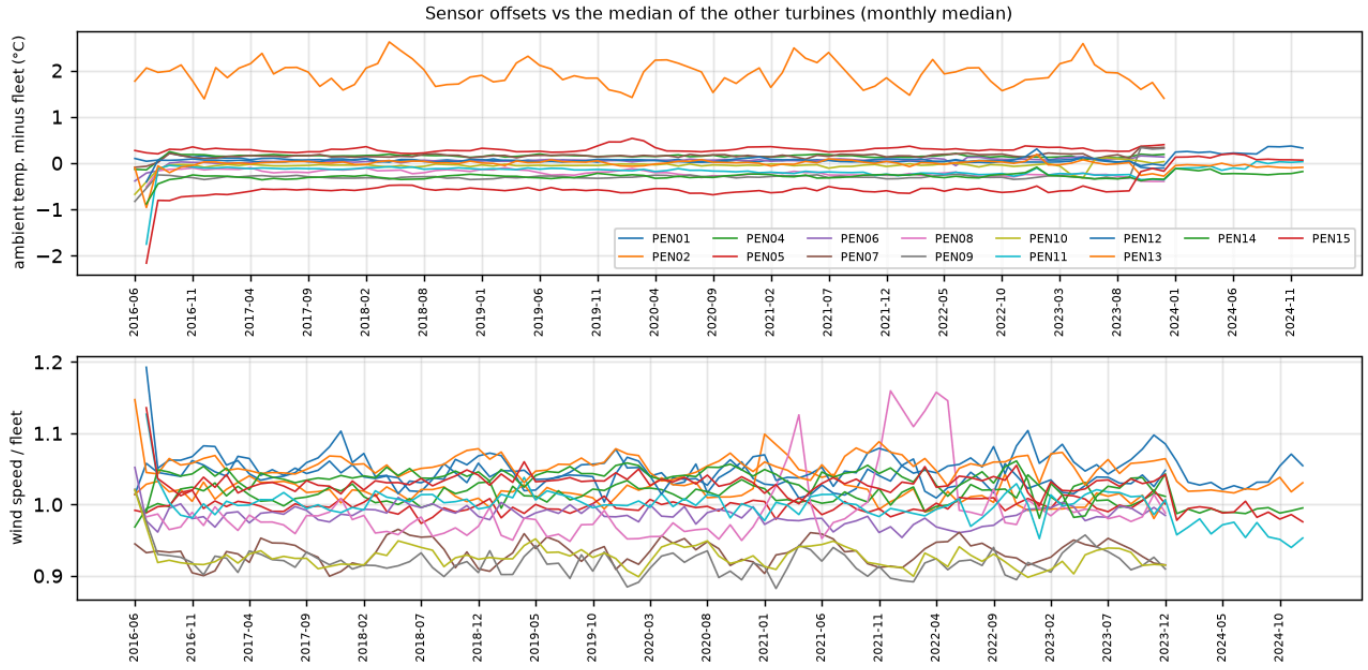
Turbine	Pump pressure (bar): baseline → last month	Inlet pressure (bar)	Oil temperature (°C)	Baseline months
PEN01	7.2 → 7.5 (+0.3)	2.5 → 2.7 (+0.2)	57.0 → 56.9 (-0.1)	2016-06..2021-12
PEN02	7.7 → 8.1 (+0.4)	2.8 → 3.0 (+0.1)	55.8 → 55.7 (-0.1)	2016-06..2021-12
PEN04	7.7 → 8.0 (+0.3)	2.9 → 3.1 (+0.3)	56.6 → 56.6 (+0.0)	2016-06..2021-12
PEN05	7.8 → 8.0 (+0.2)	2.8 → 2.9 (+0.1)	56.1 → 56.2 (+0.1)	2016-06..2021-12
PEN06	7.5 → 7.7 (+0.2)	2.8 → 2.8 (+0.0)	56.1 → 56.1 (+0.0)	2016-06..2021-12
PEN07	7.0 → 7.8 (+0.8)	2.3 → 2.6 (+0.3)	56.0 → 54.3 (-1.7)	2016-06..2021-12
PEN08	6.8 → 6.9 (+0.1)	2.5 → 2.6 (+0.1)	57.4 → 57.4 (+0.0)	2016-08..2021-12
PEN09	6.8 → 6.9 (+0.1)	2.5 → 2.6 (+0.1)	57.2 → 57.4 (+0.2)	2016-08..2021-12
PEN10	6.7 → 7.0 (+0.3)	2.5 → 2.6 (+0.1)	57.2 → 57.5 (+0.3)	2016-09..2021-12
PEN11	6.8 → 6.6 (-0.2)	2.5 → 2.4 (-0.0)	57.4 → 57.6 (+0.1)	2016-08..2022-12
PEN12	7.1 → 7.6 (+0.5)	2.6 → 2.6 (+0.0)	57.2 → 57.3 (+0.1)	2016-08..2022-12
PEN13	7.3 → 7.6 (+0.4)	2.4 → 2.6 (+0.2)	55.4 → 54.2 (-1.3)	2016-08..2022-12
PEN14	7.8 → 9.9 (+2.1)	2.6 → 2.0 (-0.6)	53.2 → 51.1 (-2.1)	2016-08..2022-12
PEN15	7.4 → 6.9 (-0.5)	2.4 → 2.3 (-0.1)	55.6 → 56.4 (+0.8)	2016-08..2022-12


Monthly medians at full load, flagged turbines highlighted (others grey)
PEN01
PEN02
PEN04
PEN05
PEN06
PEN07
PEN08
PEN09
PEN10
PEN11
PEN14
PEN15
gear-oil pump pressure (bar)
gear-oil inlet pressure (bar)
gear-oil temperature (°C)
2016-06
2016-11
2017-04
2017-09
2018-02
2018-07
2018-12
2019-05
2019-10
2020-03
2020-08
2021-01
2021-06
2021-11
2022-04
2022-09
2023-02
2023-07
2023-12
2024-05
2024-10

5. Sensor drift and stuck sensors

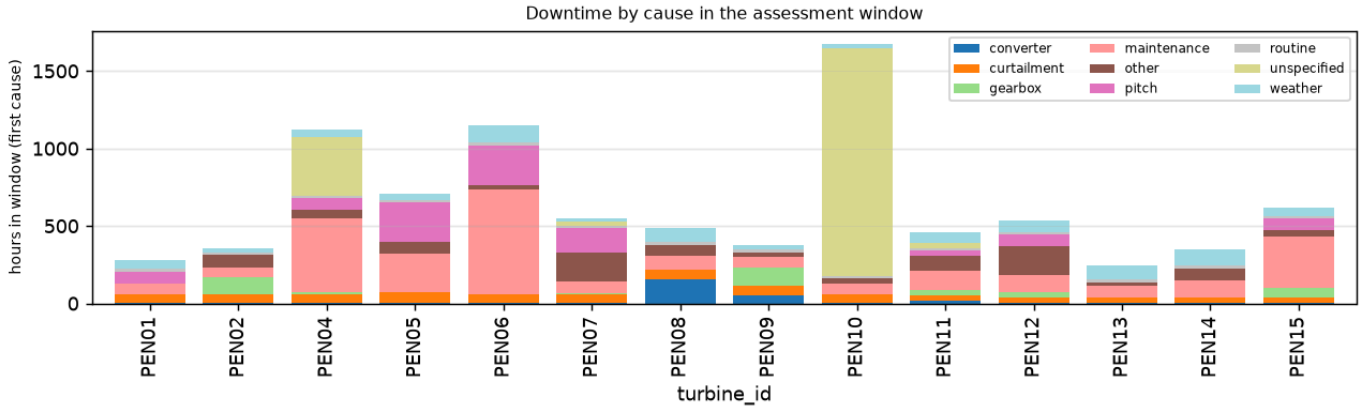
Drift = change of a turbine's offset to the median of the other turbines, last 3 months vs before the window (thresholds: 1.5 °C, 5% wind speed).

Turbine	drift ambient_temp_c	drift wind_speed_ms	Stuck sensors
PEN01	-0.17 °C	0.1%	-
PEN02	-0.36 °C	-1.5%	-
PEN04	+0.03 °C	-0.1%	-
PEN05	+0.10 °C	0.9%	-
PEN06	+0.10 °C	-0.1%	-
PEN07	+0.19 °C	-1.6%	-
PEN08	-0.20 °C	1.7%	-
PEN09	+0.62 °C	-0.3%	-
PEN10	+0.08 °C	-1.1%	-
PEN11	+0.17 °C	-5.0%	-
PEN12	+0.29 °C	0.3%	-
PEN13	-0.10 °C	-2.5%	-
PEN14	+0.06 °C	-4.9%	-
PEN15	+0.66 °C	-5.2%	-



6. Stop and fault history

Turbine	Fault stops	Maintenance	Grid	Weather	Curtailment	Major-fault episodes	Maint. h/yr	Oil / pitch trips
PEN01	24	45	1	102	15	2	34	2 / 12
PEN02	90	44	1	93	15	2	33	16 / 0
PEN04	89	44	0	80	15	4	236	12 / 2
PEN05	78	45	0	104	15	1	122	1 / 2
PEN06	54	49	1	111	16	3	336	4 / 9
PEN07	47	35	1	84	15	4	40	8 / 17
PEN08	205	46	2	115	14	7	44	9 / 3
PEN09	51	47	3	92	16	3	32	25 / 0
PEN10	25	34	27	94	16	1	35	0 / 4
PEN11	45	43	36	167	13	3	59	10 / 2
PEN12	99	46	4	173	14	2	56	6 / 3
PEN13	21	36	15	161	14	0	37	0 / 0
PEN14	33	38	3	196	14	0	56	0 / 0
PEN15	105	52	4	132	13	6	165	34 / 6



Most frequent fault stops in the window (all turbines):

Code	Message	Component	Stops
6530	Anemometer defect	sensor_met	269
4530	Tower oscillation Y level 2	tower	154
1550	Missing gear oil (high rpm)	gearbox	95
4510	Tower oscillation Y level 1	tower	91
3000	Frequency converter not ready	converter	86
3110	Frequency converter error	converter	45
3555	Current asymmetry	unspecified	31
4540	Tower oscillation X level 2	tower	19
1510	Low gearbox oil pressure	gearbox	18
4520	Tower oscillation X level 1	tower	16
650	Pitch controller communication error	pitch	15
715	Charging circuit pitch	pitch	11

7. Methods, thresholds and limits

- **Pipeline.** `windml report` reads the raw export only through the mapping (`windml.ingest`), cleans and labels each turbine (QC flags, short-gap interpolation, alarm-log stop labels), trains the models, computes the metrics and writes this report. Every number here is reproducible from the mapping and the raw files.
- **Window.** The last 24 months of each turbine; *final year* = its last 12 months.
- **Models.** LightGBM per turbine and signal on healthy operating rows (no stop within 6 h, no fault, not in the 30 days before a major fault, QC clean). With 2+ calendar years, each year is scored by a model trained only on earlier years (3-day embargo, training must cover all 12 months). With 12-23 months, the record is cut into 6 blocks scored by models trained on the other blocks (cross-fit). Under 12 months, no model.
- **RAG rules** (fixed in advance, heuristics, not a calibrated risk): RED for availability < 95%, >= 8 oil or pitch trip episodes, pump pressure >= 2 bar above baseline, or abnormal oil state on >= 30% of final-year days; AMBER for availability < 97%, >= 4 episodes, pump rise >= 1 bar, abnormal oil days >= 10%, oil-temperature model error >= 1.5x its pre-window level, or > 2% missing data; AMBER only for under-performance on >= 20% of final-year days, abnormal main-bearing or pitch residuals on >= 30%, sensor drift (1.5 °C, 5% wind) or a stuck monitored sensor.
- **Confidence.** Points for history (3+ years: 2, 2-3 years: 1), coverage >= 95%, models trained on earlier years, an alarm log, and (for flags) agreement of independent indicators. High >= 4, Medium 2-3, Low <= 1.
- **Limits.** The alarm-log classes come from the mapping's rules; trip detection uses Senvion-style oil/pitch codes and messages (other vendors need `class_rules`). Full-load oil medians assume one oil circuit design across the fleet. Models learn whatever was normal in their training data, including a fault that was already present. Sensor drift is relative to the other turbines, so a fleet-wide drift is invisible.
- **Data.** CC BY 4.0, Cubico Sustainable Investments, doi:10.5281/zenodo.5946808 (H1-2021) / 10.5281/zenodo.16807304.